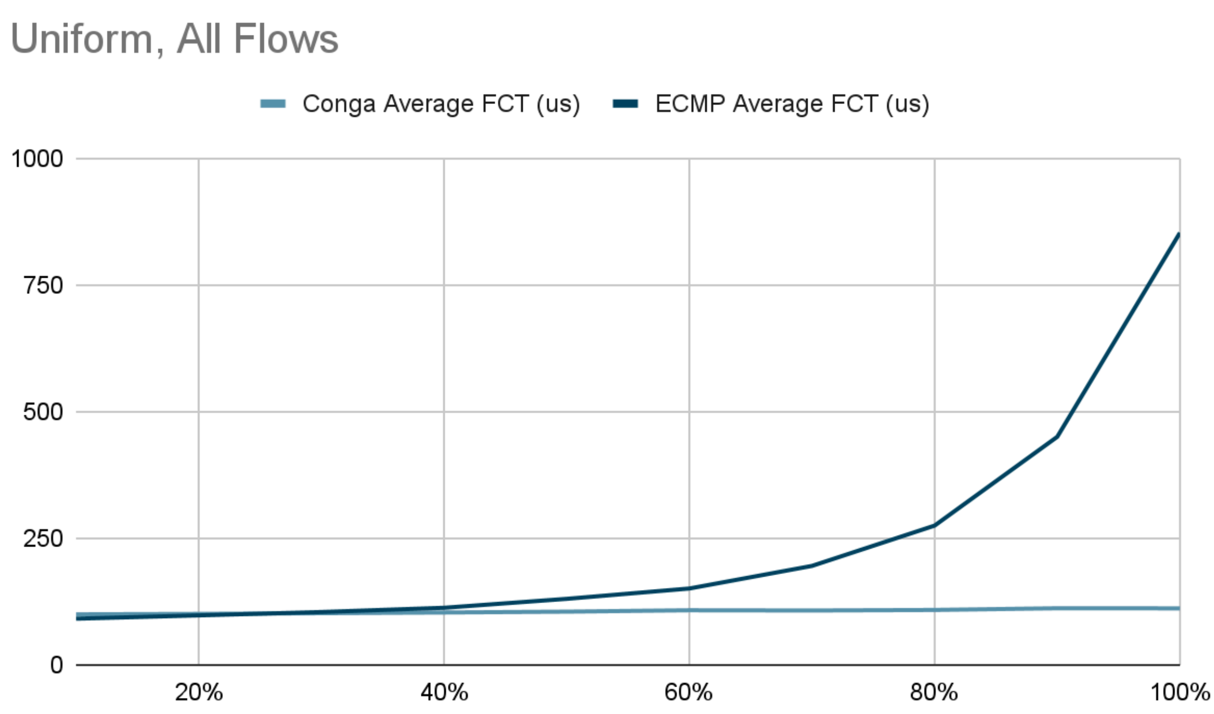
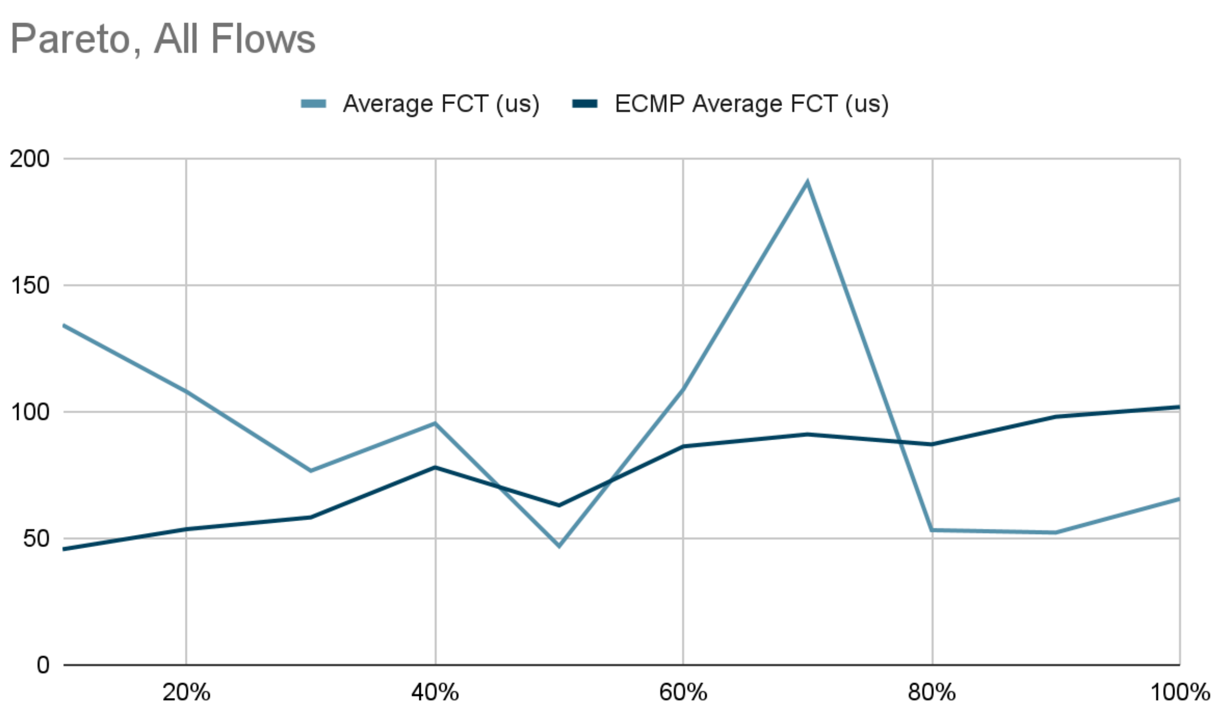
Uniform, All Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	99.91403	91.31267
20%	100.8176	97.84251
30%	102.1187	104.4912
40%	103.5665	112.8465
50%	105.2212	130.483
60%	107.9791	150.8877
70%	107.5088	195.4729
80%	108.5516	275.0342
90%	111.9278	450.271
100%	111.7514	852.8501



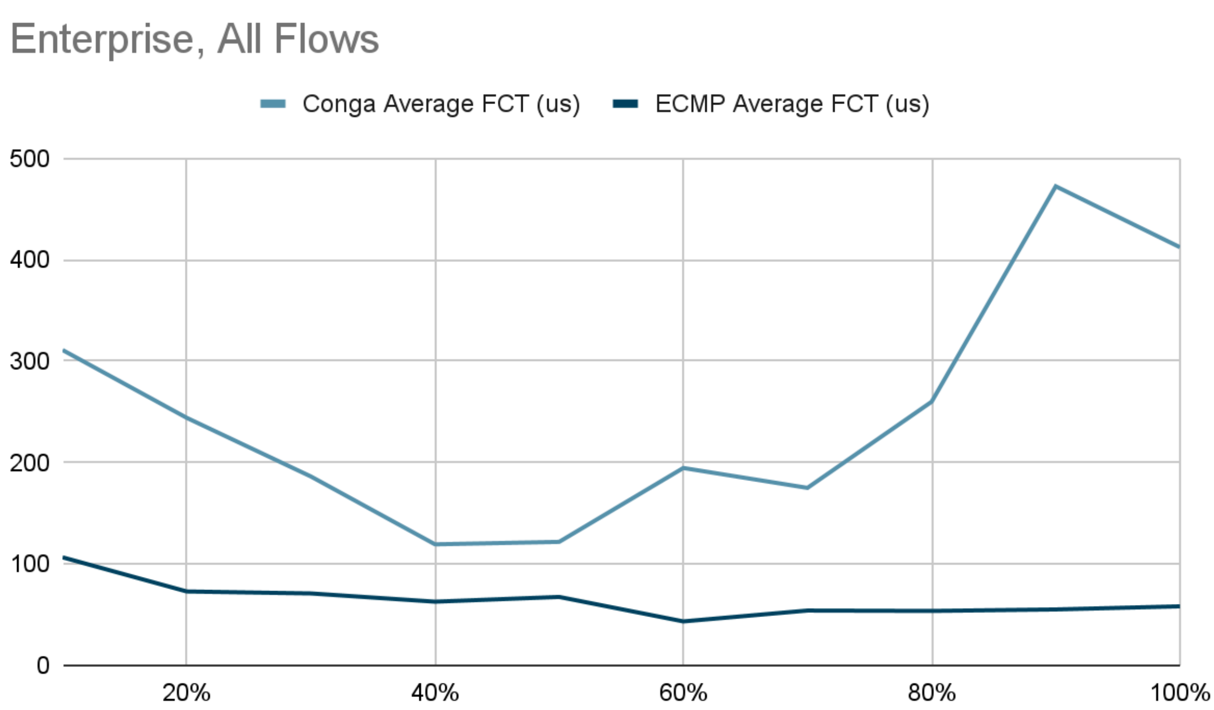
Pareto, All Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	134.1877	45.65631
20%	107.8072	53.59219
30%	76.63396	58.25122
40%	95.31537	77.9916
50%	46.91307	62.98779
60%	108.7342	86.25331
70%	190.5019	91.01278
80%	53.21735	87.05146
90%	52.24761	97.95939
100%	65.5336	101.8201



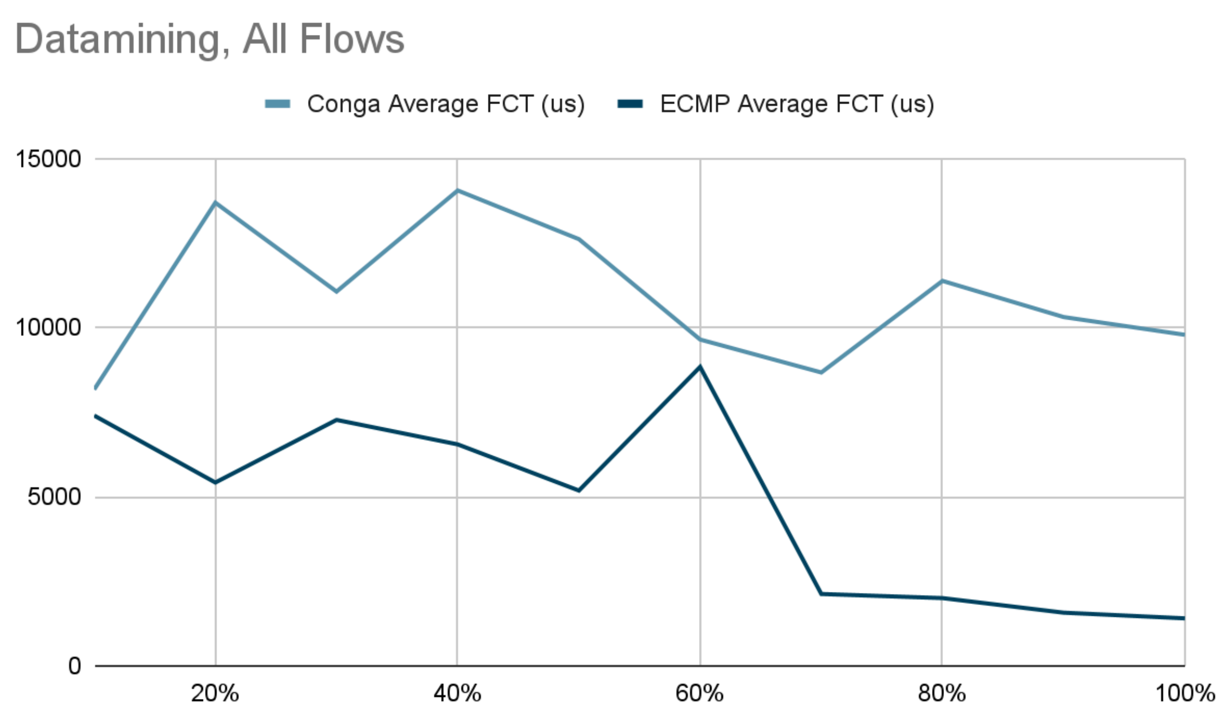
Enterprise, All Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	310.9935	106.58
20%	244.1425	72.93314
30%	186.3148	70.98654
40%	119.406	62.85766
50%	121.8401	67.53617
60%	194.6906	43.41002
70%	175.0684	54.11989
80%	259.9943	53.72717
90%	472.4339	55.20514
100%	412.3655	58.27447



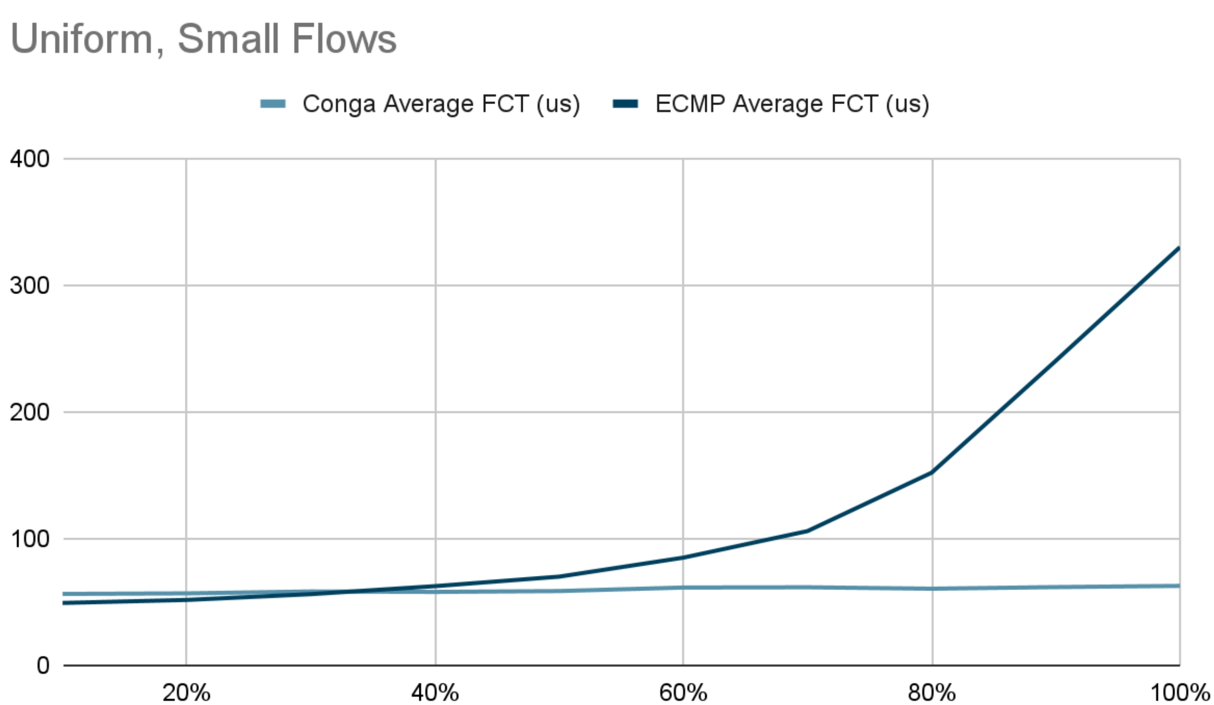
Datamining, All Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	8170.12	7400.812
20%	13692.21	5420.643
30%	11064.81	7271.82
40%	14053.41	6547.964
50%	12614.58	5183.291
60%	9647.233	8838.688
70%	8673.595	2125.6
80%	11380.77	2004.139
90%	10311.84	1572.99
100%	9787.204	1405.391



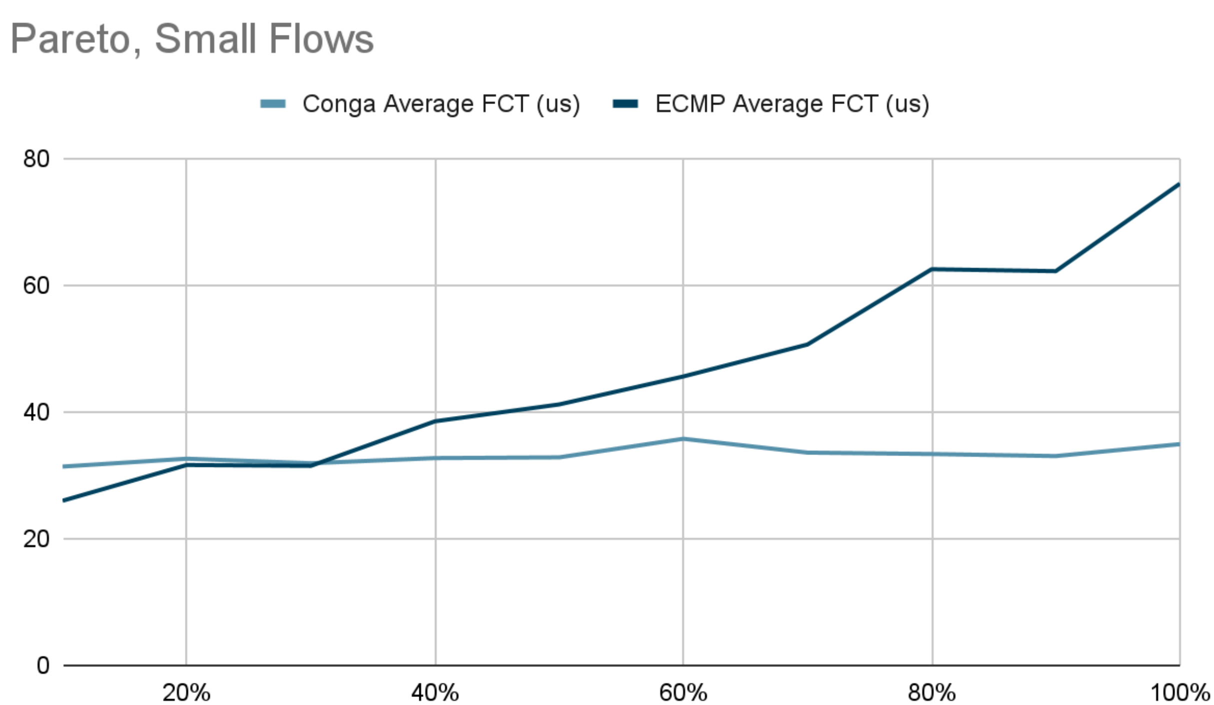
Uniform, Small Flows (Average 50KB)

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	56.40162	49.29232
20%	56.84422	51.6687
30%	58.34498	56.23724
40%	57.99634	62.53295
50%	58.66888	70.04051
60%	61.44186	85.01493
70%	61.64434	105.961
80%	60.52621	152.0232
90%	61.79214	240.3036
100%	62.75641	329.9251



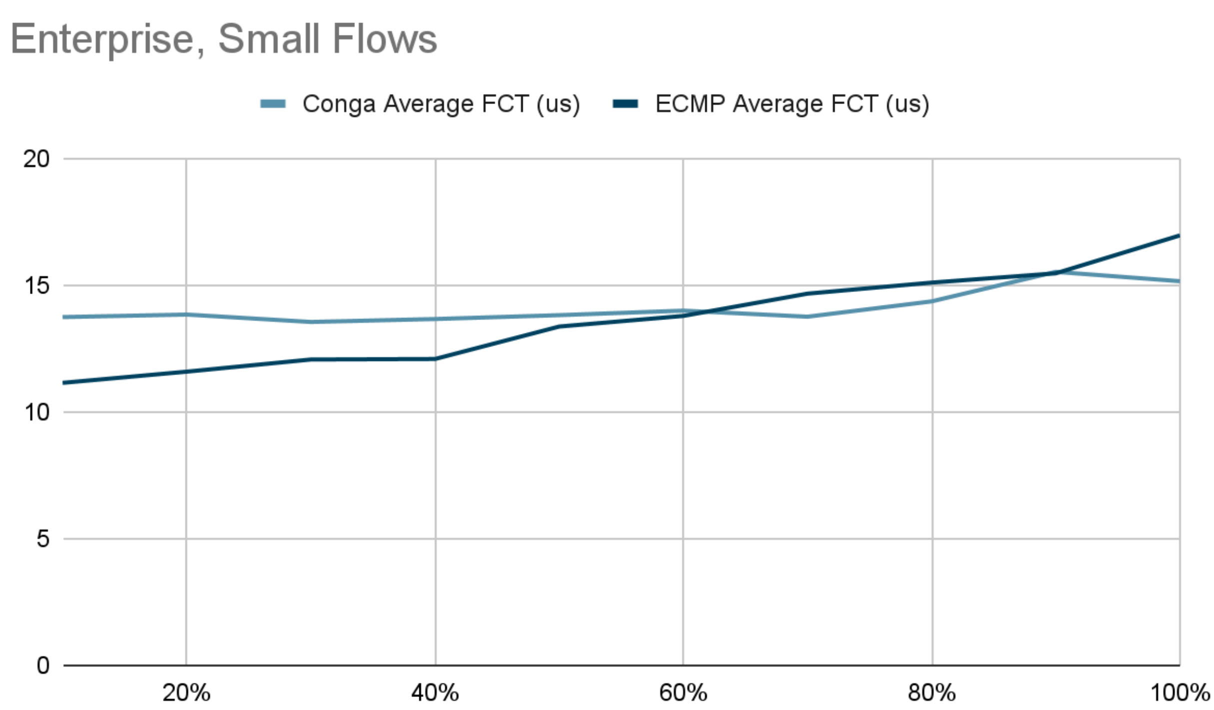
Pareto, Small Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	31.37375	25.98267
20%	32.6153	31.62034
30%	31.89849	31.48668
40%	32.71227	38.51613
50%	32.82701	41.17503
60%	35.7568	45.58884
70%	33.57597	50.62863
80%	33.35369	62.50593
90%	33.02952	62.2014
100%	34.91068	76.00815



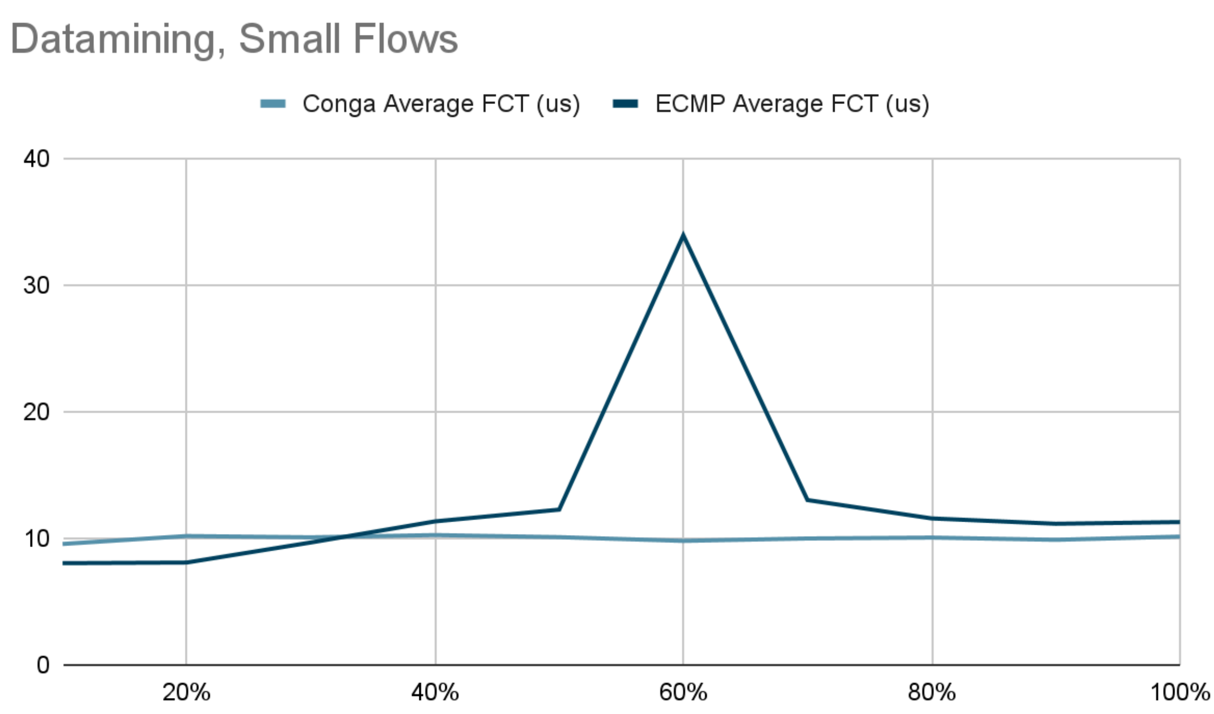
Enterprise, Small Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	13.7392	11.14526
20%	13.83851	11.58679
30%	13.54673	12.066
40%	13.66273	12.08652
50%	13.81493	13.36587
60%	13.9931	13.78462
70%	13.75396	14.66351
80%	14.35854	15.09865
90%	15.52267	15.46543
100%	15.154	16.96073



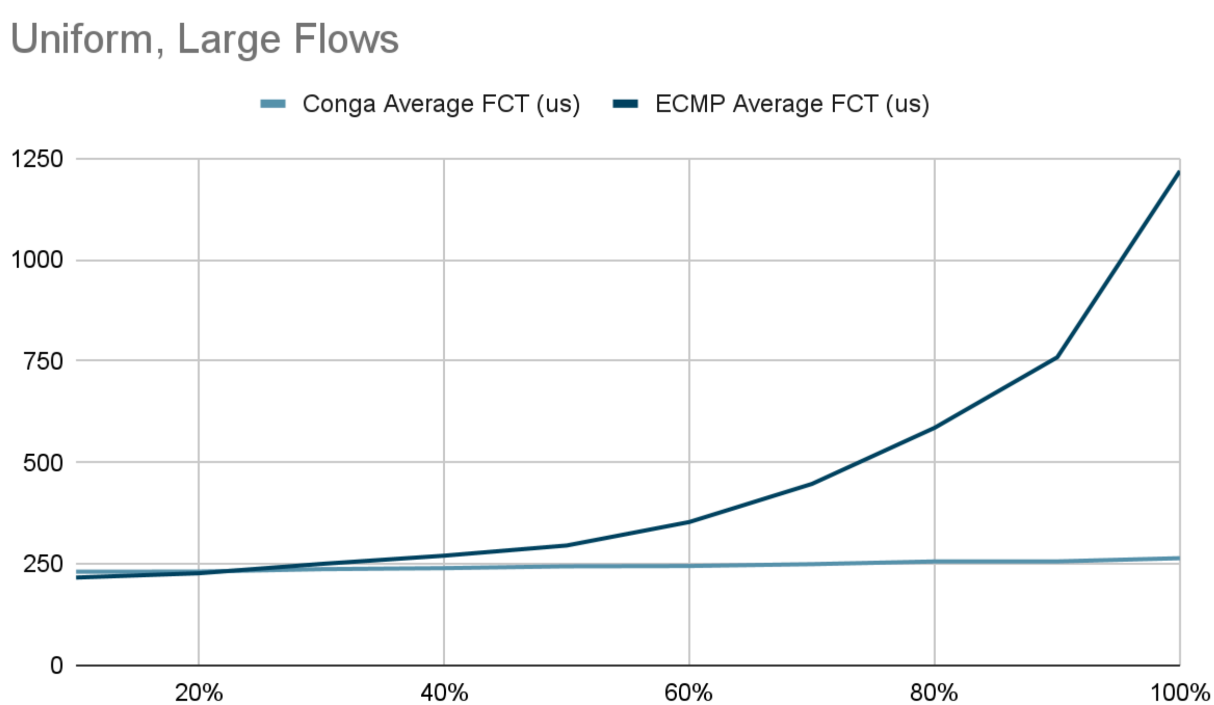
Datamining, Small Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	9.550877	8.034394
20%	10.17838	8.086578
30%	10.07583	9.668464
40%	10.25483	11.33412
50%	10.08869	12.26084
60%	9.793251	33.91344
70%	9.9848	13.01843
80%	10.0554	11.56765
90%	9.877624	11.14601
100%	10.13224	11.28103



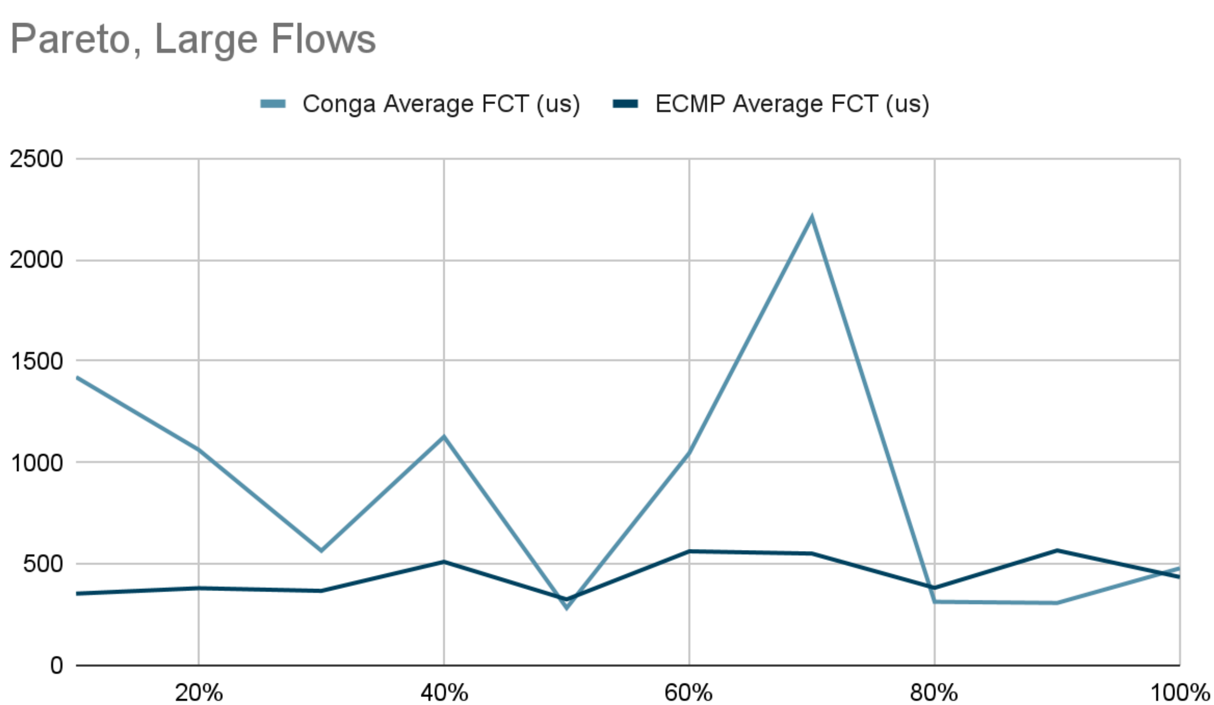
Uniform, Large Flows (Average 250KB)

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	230.9666	216.6268
20%	231.4078	227.1773
30%	237.323	250.2733
40%	239.8359	270.5042
50%	244.3102	295.6489
60%	245.1311	353.5187
70%	249.6471	447.4852
80%	256.3168	585.8105
90%	256.4109	759.6094
100%	264.3752	1218.802



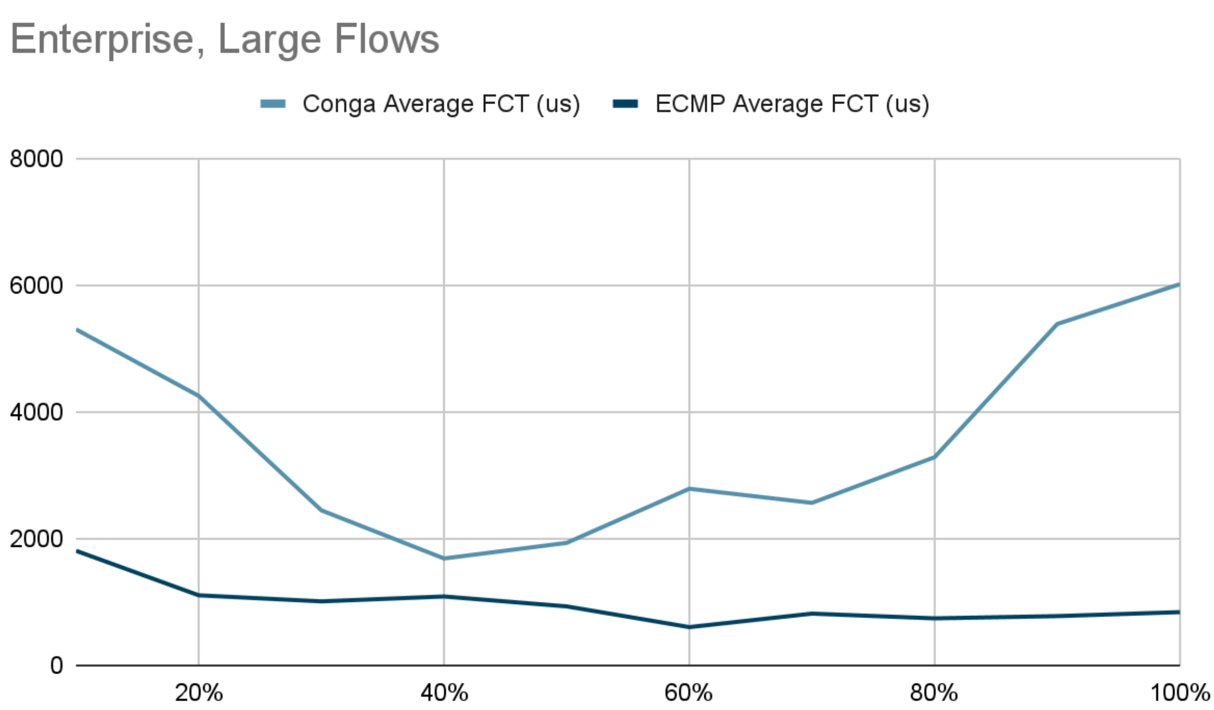
Pareto, Large Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	1421.418	353.9842
20%	1062.947	380.7669
30%	565.5663	367.5303
40%	1127.187	510.8286
50%	283.4105	325.5799
60%	1047.444	562.5359
70%	2209.042	551.4906
80%	313.652	382.5419
90%	308.0377	566.7865
100%	478.943	435.6072



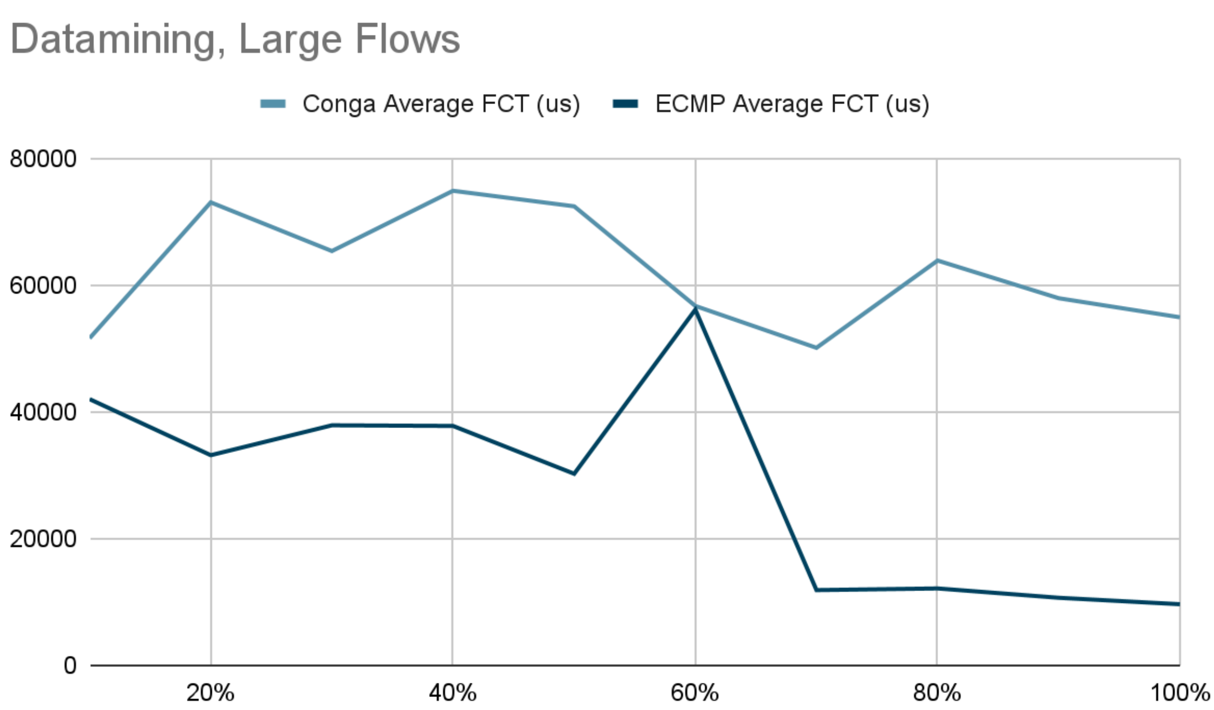
Enterprise, Large Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	5301.737	1808.499
20%	4251.431	1106.135
30%	2446.697	1010.769
40%	1687.381	1087.8
50%	1933.631	931.5524
60%	2787.426	604.5499
70%	2564.539	817.2062
80%	3285.193	742.6025
90%	5386.886	778.0179
100%	6014.826	840.8054



Datamining, Large Flows

Load	Conga Average FCT (us)	ECMP Average FCT (us)
10%	51623.85	42010.58
20%	73053.41	33168.15
30%	65374.38	37884.58
40%	74882.45	37782.65
50%	72432.53	30236.04
60%	56713.8	56129.74
70%	50110.4	11883.16
80%	63875.57	12145.97
90%	57929.77	10671.52
100%	54924.69	9656.247



Evaluation

My CONGA implementation underperforms ECMP because of poor handling for large flows (likely erroneous implementation somewhere). This is most evident with the relatively strong performance of the small flows. However, there were also difficulties with data gathering as the data was copied from the powershell output instead of using the trace logs which meant that some statistics from the simulator runs were missed.
The only flow distribution in which my CONGA implementation consistently outperforms ECMP is in uniform flow distribution.

REFERENCES:
Implementation aided with AI Coding tool: Github Copilot